

Junran Wang

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EDUCATION

2025 - Present	Georgia Institute of Technology Master of Science in <i>Electrical and Computer Engineering</i>	Atlanta, GA GPA: 4.0/4.0
2021 - 2025	Tsinghua University Bachelor of Engineering in <i>Vehicle Engineering (Electronic Information)</i>	Beijing, China GPA: 4.0/4.0
2024	University at Buffalo, the State University of New York Visiting Scholar in <i>Department of Computer Science and Engineering</i>	Buffalo, NY

RESEARCH EXPERIENCE

SEAD: A State-Based Perspective on Attack and Defense in Tool-Using Agents Jul 2026 - Present
Georgia Institute of Technology | Co-first Author |  Website Advisor: Dr. Pan Li

- Formulated attack and defense for tool-using agents as **partially observed state control**, grounding safety in persistent environment changes and harmful-state reachability.
- Developed DART and SAGE for adversarial evaluation and **state-aware pre-execution defense**, supported by re-playable tool environments and task-specific executable checks.
- SAGE preserved **95.79%** of benign trajectories and intercepted **92.73%** of harmful paths in recorded evaluations; reduced executable attack success from **48.0% to 4.0%** in online evaluation.

AcFlow: Controlling T2I DiTs via Learned Conditional Activation Flow Jul 2026 - Sept 2026
Georgia Institute of Technology | First Author |  Website

- Introduced ACFlow, a learned concept-conditioned **velocity field** that steers image-token activations in frozen text-to-image diffusion transformers through **token-specific, activation-dependent updates**.
- Enabled **continuous** style control and concept suppression through adjustable flow horizons, with weights **shared across concepts** within each task family and **zero-shot generalization**.
- Achieved the **best style-content trade-off** among evaluated baselines.

ImmersedPrivacy: Evaluating VLMs' Privacy Awareness in the Physical World Jan 2026 - May 2026
Georgia Institute of Technology | First Author |  Website Advisor: Dr. Pan Li

- Developed IMMERSEDPRIVACY, a first-of-its-kind **interactive Visual+Audio benchmark** to evaluate VLMs' privacy awareness in dynamic physical environments.
- Engineered a three-tier hierarchical framework built on a customized Unity simulator featuring 53 complex scenarios and 500+ procedurally generated instances.
- Implemented an **interactive evaluation pipeline**, enabling agents to actively navigate environments and resolve task-privacy conflicts through multi-turn, closed-loop decision making.

Beyond Steering Vector: Flow-based Activation Steering Mar 2026 - May 2026
Georgia Institute of Technology | Co-first Author |  Website Advisor: Dr. Chao Zhang

- Introduced FLAS, a concept-conditioned velocity field that steers frozen LLM activations through adaptive, multi-step, and position-sensitive flows using positive examples alone.
- Designed a lightweight FlowBlock architecture with cross-attention and Euler integration, enabling zero-shot control of unseen concepts and continuous steering-strength adjustment without per-concept tuning.
- Achieved held-out HMean scores of **1.015/1.113** on Gemma-2-2B/9B-IT, making FLAS the first learned steering method to consistently outperform in-context prompting on AxBench.

WORK EXPERIENCE

Machine Learning Engineer Intern Sept 2024 - May 2025
Infplane AI Ltd.

- Developed automated hyperparameter optimization algorithms to reduce training costs for LLMs.
- Developed a non-supervision graph-based algorithm for root-cause analysis of a computation cluster.

PUBLICATIONS

Shen, *Xinjie, *Junran Wang, Rongzhe Wei, and Pan Li (2026). *SEAD: A State-Based Perspective on Attack and Defense in Tool-Using Agents*. arXiv: [2609.34518](https://arxiv.org/abs/2609.34518) [cs.CR]. URL: <https://arxiv.org/abs/2609.34518v1>.

Wang, *Junran, *Zehao Jin, Tianyu Luan, and Xinjie Shen (2026). *AcFlow: Controlling Text-to-Image Diffusion Transformers via Learned Conditional Activation Flow*. arXiv: [2609.10723](https://arxiv.org/abs/2609.10723) [cs.CV]. URL: <https://arxiv.org/abs/2609.10723>.

*Wang, Junran, *Xinjie Shen, *Zehao Jin, and Pan Li (2026). “How far are VLMs from privacy awareness in the physical world? an empirical study”. In: *NeurIPS 2026*. URL: <https://arxiv.org/pdf/2605.05340>.

*Jin, Zehao, *Ruixuan Deng, *Junran Wang, Xinjie Shen, and Chao Zhang (2026). “Beyond Steering Vector: Flow-based Activation Steering for Inference-Time Intervention”. In: *NeurIPS 2026*. URL: <https://arxiv.org/pdf/2605.05892>.

Huang, Yinan, Hans Hao-Hsun Hsu, Junran Wang, Bo Dai, and Pan Li (2026). “Accelerated Sequential Flow Matching: A Bayesian Filtering Perspective”. In: *ReALM-GEN 2026 - ICLR 2026 Workshop*. URL: <https://arxiv.org/pdf/2510.02356>.

HONORS & AWARDS

National Scholarship	Top 2% university students in China	2024
Overall Excellence Scholarship	Top 5% in Tsinghua University	2023
Finalist in Interdisciplinary Contest in Modeling (ICM)	Top 2% teams worldwide	2023
Overall Excellence Scholarship	Top 5% in Tsinghua University	2022

SKILLS

Programming Languages	C, C++, Python, familiar with RUST
ML Frameworks & Serving	PyTorch, CUDA, Hugging Face Transformers, vLLM, SGLang
AI Agent Infra	Post-training, MCP, MCP, Trajectory Analysis
Software Engineering	Docker, Docker Compose, Slurm, Git
Rendering	Unity Engine, Blender, Meshlab
Languages	Mandarin (native), English (TOEFL 115), Italian (entry-level)